Example 7.18

Prove that the ellipse $x^2+4y^2=8$ and the hyperbola $x^2-2y^2=4$ intersect orthogonally.

Solution

Let the point of intersection of the two curves be (a,b). Hence,

$$a^2+4b^2 = 8 \text{ and } a^2-2b^2=4 \qquad \ldots (4)$$

It is enough to show that the product of the slopes of the two curves evaluated at (a,b) is -1.

Differentiation of $x^2+4y^2=8$ with respect x, gives

$$2x+8y\frac{dy}{dx} = 0$$

$$\text{Therefore } \frac{dy}{dx} = -\frac{x}{4y}.$$

$$\text{Then, } \frac{dy}{dx} \text{ at } (a,b) = m_1 = -\frac{a}{4b}.$$

Differentiation of $x^2-2y^2=4$ with respect to x, gives

$$2x-4y\frac{dy}{dx} = 0$$

$$\text{Therefore, } \frac{dy}{dx} = \frac{x}{2y}.$$

$$\text{Then } \frac{dy}{dx} \text{ at } (a,b) = m_2 = \frac{a}{2b}.$$

Therefore,
$$m_1 \times m_2 = \left(-\frac{a}{4b}\right)\times\left(\frac{a}{2b}\right) = -\frac{a^2}{8b^2} \qquad \ldots (5)$$

Applying the ratio of proportions in (4), we get

$$\frac{a^2}{-16-16} = \frac{b^2}{-8+4} = \frac{1}{-2-4}.$$

Therefore $\frac{a^2}{b^2} = \frac{32}{4} = 8$. Substituting in (5), we get $m_1 \times m_2 = -1$. Hence, the curves cut orthogonally. ■

EXERCISE 7.2

1. Find the slope of the tangent to the following curves at the respective given points.

   (i) $y = x^4+2x^2-x$ at $x=1$ (ii) $x = a\cos^3 t, y = b\sin^3 t$ at $t=\frac{\pi}{2}$.

2. Find the point on the curve $y = x^2-5x+4$ at which the tangent is parallel to the line $3x+y=7$.

3. Find the points on the curve $y = x^3-6x^2+x+3$ where the normal is parallel to the line $x+y=1729$.

4. Find the points on the curve $y^2-4xy = x^2+5$ for which the tangent is horizontal.

Chapter 7 Differential Calculus Original-new.indd 14 26-02-2020 17:22:29